

AUTOMATED PRODUCE SCANNER

Bill of Materials

Real-Time Pricing and Packaging System

Portfolio / Technical Documentation Edition

1. Bill of Materials Overview

The Bill of Materials (BOM) identifies the hardware, electronic components, mechanical materials, and supporting accessories used in the development of the Automated Produce Scanner prototype. The list is based directly on the Bill of Materials presented in the submitted capstone thesis.

The prototype combines a Raspberry Pi 4 Model B, camera module, load-cell weighing system, LCD, thermal printer, barcode scanner, actuator-driven sealing mechanism, power components, and mechanical materials. The thesis reports a total materials cost of ₱19,411.00.

2. Complete Bill of Materials

Component	Quantity	Unit Cost	Total Cost	Function / Note
Raspberry Pi 4 Model B	1	₱6,175.00	₱6,175.00	Main processing unit and system controller
Load Cell with HX711 Module	1	₱435.00	₱435.00	Weight measurement and signal conversion
Raspberry Pi Camera Module V2 (IMX219)	1	₱1,935.00	₱1,935.00	Image acquisition for produce detection
Liquid Crystal Display with I²C Module (LCD)	1	₱70.00	₱70.00	Displays produce, accuracy, weight, and price
Push Button	2	₱13.00	₱26.00	User input/control
Resistor (330 Ω)	1	₱2.00	₱2.00	Electronic support component
Linear Actuator	1	₱1,680.00	₱1,680.00	Drives the automated sealing mechanism
Heat Impulse Sealer	1	₱580.00	₱580.00	Seals heat-sealable packaging
Power Supply	1	₱415.00	₱415.00	Provides electrical power
Barrel Jack	1	₱130.00	₱130.00	Power connection
Stepper Motor Drive	1	₱340.00	₱340.00	Listed hardware component; not part of the main implemented setup
Buck Converter	1	₱85.00	₱85.00	Voltage conversion/regulation

Terminal Block (Set)	20	₱44.00 per 10 pieces	₱88.00	Organizes and secures wiring connections
RPi Flex Cable (1000 mm)	1	₱130.00	₱130.00	Camera-to-Raspberry Pi connection
Thermal Printer	1	₱1,595.00	₱1,595.00	Prints barcode/transaction labels
Thermal Sticker	1	₱210.00	₱210.00	Printing material for labels
Barcode Scanner	1	₱870.00	₱870.00	Scans printed transaction barcode
Angle Bar (Stainless)	15	₱5.00	₱205.00	Mechanical structural material
Flat Head Screw (Set)	48	₱205.00	₱205.00	Mechanical fastening
Acrylic Casing	1	₱3,600.00	₱3,600.00	Main enclosure/structural casing
Plywood (10 mm)	1	₱600.00	₱600.00	Mechanical/platform material
Pulley	2	₱17.50	₱35.00	Mechanical transmission component
TOTAL			₱19,411.00	Total materials cost reported in the thesis

Total Materials Cost: ₱19,411.00

3. BOM by Functional Category

Category	Components
Processing and Computing	Raspberry Pi 4 Model B
Sensing and Computer Vision	Raspberry Pi Camera Module V2 (IMX219); Load Cell with HX711 Module
User Interface and Controls	LCD with I²C Module; Push Buttons; Resistor (330 Ω)

Transaction and Identification	Thermal Printer; Thermal Sticker; Barcode Scanner
Packaging and Actuation	Linear Actuator; Heat Impulse Sealer; Pulley
Power and Electrical Distribution	Power Supply; Barrel Jack; Buck Converter; Terminal Block; RPi Flex Cable
Mechanical Structure	Acrylic Casing; Plywood (10 mm); Stainless Angle Bar; Flat Head Screw Set
Listed / Future Mechanical Component	Stepper Motor Drive — listed in the thesis BOM but explicitly noted as not part of the main implemented setup

4. Core System Components

For a technical portfolio, the following components are the most important to highlight because they directly support the core functionality of the system:

- Raspberry Pi 4 Model B — central processing and system control.
- Raspberry Pi Camera Module V2 — image acquisition for produce detection.
- Edge Impulse deployment — machine-learning inference software.
- Load Cell + HX711 — weight measurement.
- LCD with I2C — real-time user display.
- Thermal Printer + Thermal Sticker — transaction/barcode output.
- Linear Actuator + Heat Impulse Sealer — automated packaging.
- Local database — produce pricing and transaction information.

5. Source Accuracy Notes

The quantities, unit costs, total costs, and overall total in this document preserve the values shown in the submitted thesis. The listed total costs add up to the reported ₱19,411.00 total.

The thesis lists a Stepper Motor Drive in the Bill of Materials but separately indicates that it was not part of the main implemented setup and may be considered for future mechanical control. For a GitHub portfolio, it is therefore better to label this component as 'listed / future' rather than implying that it was a core operating component.

The BOM should be treated as the cost record for the original academic prototype. Current replacement prices may differ and should not be presented as current market prices without separate verification.

6. Recommended GitHub Presentation

For the public GitHub version, a shortened BOM is preferable to the full academic cost table. A recruiter primarily needs to understand what technologies and hardware were integrated.

Recommended README table:

Hardware	Purpose
Raspberry Pi 4 Model B	Central processing
Raspberry Pi Camera Module V2	Computer vision input
Load Cell + HX711	Weight measurement
LCD + I²C	User interface
Thermal Printer	Barcode/transaction output
Barcode Scanner	Transaction retrieval
Linear Actuator	Packaging actuation
Impulse Sealer	Automated packaging
Power Supply / Buck Converter	Power management

7. Documentation Note

This document is a portfolio-oriented adaptation of the Bill of Materials from the submitted capstone thesis. It is intended to document the hardware used in the prototype without reproducing the full thesis. The original thesis is identified as an unpublished academic capstone project.